Energy Savings From Advanced Process Control: Two Case Studies

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ABSTRACT

As energy costs continue to increase and effluent permits become more stringent, process control has become one of the most important components of the operations of activated sludge processes. Utilizing a process controller that adjusts the dissolved oxygen (DO) set points based on real time influent loading can result in significant energy savings at a biological nutrient removal wastewater treatment plant (WWTP). This paper presents two case studies of plants that implemented such a control system. At the Metropolitan Wastewater Reclamation District's Robert W. Hite Treatment Facility in Denver, Colorodo, an average of 11% less air was used when using the process controller compared to fixed DO set-point control. At the Liuwei WWTP, located in Yangzhou, Jiangsu Province, China, the plant achieved an energy savings of 21.0% using a process controller compared to fixed aeration.

KEYWORDS: Process Control, Process Optimization, Energy Savings

INTRODUCTION

With continuing increases in energy costs and the requirements of biological nutrient removal (BNR), process control has become one of the most important components of the operations of activated sludge processes.

The energy use of the aeration system for the activated sludge process accounts for approximately sixty percent of the total energy use of the plant (EPA, 1989). A well designed and implemented process control system can save significant amounts of energy.

The purpose for process control at wastewater treatment plants (WWTPs) is to respond to disturbances. WWTPs are continuously subject to disturbances caused by the inflow side of the process. The wastewater influent varies in flow, concentration, and composition throughout the day and year. The WWTP must still maintain consistent effluent quality despite these diurnal fluctuations. A process control system with proper instrumentation will minimize the effect of diurnal fluctuations on the effluent quality while achieving treatment goals.

Dynamic dissolved oxygen (DO) set-point control has started to become popular with the increased cost of energy and the emergence of better online instrumentation. The basic concept of dynamic DO set-point control is to vary the DO set-point with the influent loading to meet an effluent goal. Successful implementation can provide aeration energy savings of 10 to 20% while maintaining the effluent quality. There are two control principles, feedback and feed forward, that are used with dynamic DO set-point control.

There have been several case studies (Liu, 2005; Walz, 2009) of facilities using ammonium analyzers to enhance the aeration control system. This paper discusses the implementation and results of a feed forward process controller used in two energy savings projects.

METHODOLOGY

Control Theory

Wastewater treatment plants (WWTP) do not typically operate at the maximum design conditions during most of the day, most of the year, and lifetime of the plant, which provides flexibility to optimize the operations. The aeration system of an activated sludge process is the highest energy consuming system within the treatment train. With the use of DO and ammonia control the energy consumption of the aeration system can be lowered tremendously by taking advantage of the additional treatment capacity during diurnal lows.

The purpose of the DO set-point and ammonia controller is to meet effluent ammonia targets while minimizing the amount of oxygen supplied to the process. Minimizing the supplied oxygen reduces the loading on the air blowers, resulting in significant energy and monetary savings.

Residual DO in each aerobic stage of the process can be used to control the specific nitrification rate of the biomass. Increasing the DO concentration increases the nitrification rate by increasing the availability of oxygen to the autotrophic biomass. The increase in the nitrification rate results in a decrease in the effluent ammonia concentration. The converse is also true; lowering the DO concentration will increase effluent ammonia.

Feed-Forward Process Controller Description

The DO set-point controller uses a feed-forward control algorithm to match the DO set points in the aerobic stages to the variations in diurnal loading. The control algorithm utilizes a biological mathematical model that is customized to the plant and the specific process to calculate an ammonia concentration profile throughout the bioreactor. The model is used with an optimization algorithm to determine the effect different DO set points will have on the effluent ammonia concentration. The optimization algorithm ensures the DO set points calculated are the lowest possible to maintain effluent ammonia targets. Figure 1 is a diagram that visually explains the feed-forward DO set-point controller.

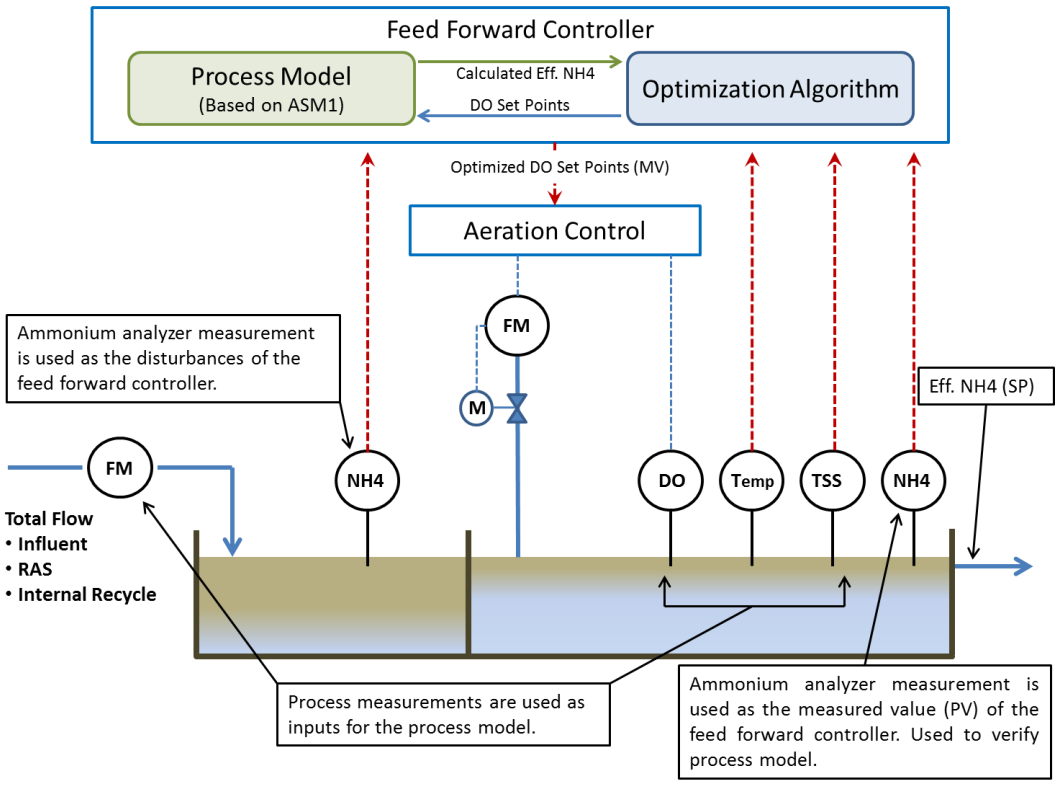

Figure 1. General Dissolved Oxygen and Ammonia Control Diagram

<u>Summary of Measured Values</u>

The feed-forward control algorithm requires the following inputs:

- Effluent Ammonia (mg/l) (End of Aerobic zone)
- Influent Ammonia (mg/l) (End of Pre-Anoxic zone)
- Dissolved Oxygen (mg/l) (one in each aeration zone)
- MLSS (mg/l) within the bioreactors
- Temperature (C˚) within the bioreactor
- Influent flow rate (MGD)
- Return Activated Sludge flow rate (MGD)
- Mixed Liquor Return flow rate (MGD)

At a minimum only one set of ammonium analyzers is required; the controller is capable of calculating the influent ammonia concentrations of multiple trains if the flows are known.

<u>Controlled Values:</u>

The residual ammonia concentration at the end of aeration is used as a control set-point for all trains. This effluent ammonia concentration target set-point is selected by operators. The ammonia set-point can be changed on the user interface at any time.

Manipulated Values:

The DO set points for the individual aerobic stages are used to control the bioreactor effluent ammonia concentration. The controller DO set-point calculations are limited to a DO set point range and maximum step change selected by the operator.

Activated Sludge Model Description

The controller utilizes a mathematical model based upon the International Water Association (IWA) Activated Sludge Model #1 (ASM1) that is customized to the plant specific process. The purpose of the model is to calculate the ammonia concentration within the bioreactor and to work with the process optimization algorithm to calculate the lowest DO set points to achieve the ammonia effluent goal. The model incorporates both the biological kinetics of the activated sludge process and physical characteristics of the specific plant and the treatment process. While it is not a complete hydraulic model, the model calibration process indirectly accounts for the impact of hydraulic heterogeneities on the treatment performance.

Biological Kinetics

The purpose of the model is to simulate the concentrations of ammonia within the aerobic reactors, so the model only incorporates process rates from ASM1 that remove or add ammonia to the process. The IWA model process rates include several parameters that are difficult to measure, making it difficult to tune. Below is the method that was used to modify the original ASM1 process rates.

$$\underbrace{-\left(f_{NH}+\frac{1}{Y_A}\right)}_{\text{Autotrophic Uptake}}\underbrace{\mu_A\cdot\frac{S_{NH}}{K_{NH}+S_{NH}}\cdot\frac{S_{DO}}{K_{DO_A}+S_{DO}}\cdot X_A}_{\text{Nitrification}}+\underbrace{k_a\cdot S_{ND}\cdot X_H}_{\text{Ammonification}}\underbrace{-f_{NH}\cdot\mu_H\cdot\frac{S_S}{K_S+S_S}\cdot\frac{S_{DO}}{K_{DO_H}+S_{DO}}\cdot X_H}_{\text{Heterotrophic Uptake}}$$

$$\underbrace{-\left(\cancel{f_{NH}}+\frac{1}{Y_A}\right)}_{\text{Insignificant}}\underbrace{\mu_A\cdot\frac{S_{NH}}{K_{NH}+S_{NH}}\cdot\frac{S_{DO}}{K_{DO_A}+S_{DO}}\cdot X_A}_{\text{Modified to use measurable variables}}\underbrace{\cancel{+k_a\cdot S_{ND}\cdot X_H}\cancel{-f_{NH}\cdot\mu_H\cdot\frac{S_S}{K_S+S_S}\cdot\frac{S_{DO}}{K_{DO_H}+S_{DO}}\cdot X_H}}_{\text{Incorporated into measured } NH_4 \text{ adjustment}}$$

$$\underbrace{SARR}_{\substack{\text{Specific Ammonia Removal Rate}\\ \text{(mg NH4 removed per g of TSS)}}}\cdot\frac{S_{NH}}{K_{NH}+S_{NH}}\cdot\frac{S_{DO}}{K_{DO_A}+S_{DO}}\cdot\underbrace{X_{MLSS}}_{\substack{\text{Mixed Liquor Suspended Solids}\\ \text{(g of TSS)}}}$$

The specific ammonia removal rate is essentially the nitrification process rate with modifications to be able to use measured values instead of the IWA model variables such as growth rate,

biomass, and yield for autotrophs. Like the IWA model, the actual ammonia removal rate is dependent upon limiting variables ammonia, and dissolved oxygen.

Feed Forward Controller Program

The feed forward program consists of two parts, the user interface and the process control engine.

The user interface is programmed and compiled utilizing the development software Visual Basic by Microsoft. The operational modes and set points can be accessed, implemented, and modified directly through the user interface. Additionally, soft-coded tuning parameters can easily be accessed through the Human Machine Interface (HMI), which allows for on-the-fly control tuning, giving plant operators and engineers the ability to adjust the system as necessary. All data necessary for controller operation is easily visible on the user interface with the train overview screens and historical graphing

The process control engine is where the process model and optimization algorithm calculations are implemented. The mathematical software MatLab is used to implement the process control engine.

CASE STUDY 1

The Metropolitan Wastewater Reclamation District, Denver, Colorodo

The Metropolitan Wastewater Reclamation District (Metro District) in Denver, Colorado, treats wastewater from Denver and parts of northern Colorado at its Robert W. Hite Treatment Facility (RWHTF), with an average daily flow of about 530,000 m^3/day (140 MGD). The RWHTF currently consists of two distinct secondary treatment plants. The South Secondary (SSEC) is a pure oxygen plant, and the North Secondary (NSEC) is an activated sludge plant. The NSEC is set up as four "quads" with three trains each. The trains are configured as a Modified-Ludzak-Ettinger (MLE) process, with three anoxic zones and one aeration zone in the first pass, and one aeration zone in each of the second and third passes. The NSEC is also equipped with centrate and return activated sludge re-aeration basins (CaRRB), but this process was not the primary focus of the study. The NSEC treats approximately 303,000 m^3/day (80 MGD).

In 2009, the Metro District spent over $4.8 million in electrical charges, of which approximately one third was for operating the blowers in the North Secondary. In 2010, the Metro District created the NSEC Energy Optimization Initiative. The initiative was formed to allow operating staff to identify and implement operational changes and technologies to reduce energy consumption.

As part of NSEC Energy Optimization Initiative, the Metro District's Model User Group (MUG) worked with consultants to conduct model simulations to calculate potential aeration savings at different DO set points. BioChem Technology (BioChem) also modeled the process to quantify the potential benefits of using feed-forward dynamic DO set point control. The simulations showed that feed-forward control was able to save 12 to 23% in aeration compared to current operating parameters without affecting the effluent quality. Based upon the simulation results,

the Metro District contracted with BioChem to demonstrate the energy savings potential of a feed forward controller.

NSEC Pilot Installation

The controller installed for the demonstration project was the Bioprocess Intelligent Operating System (BIOS) as manufactured by BioChem. The BIOS uses a feed-forward control and optimization algorithm to determine the optimal set points for DO to achieve treatment goals while minimizing energy consumption. The demonstration started on September 1, 2011 and ended in March of 2012.

For the NSEC pilot installation, the controller was programmed to control one quad (Trains 7, 9 and 11). Train 9 is instrumented with ammonium analyzers in the first anoxic zone and at the end of the last aeration basin. The instrumented train is considered the "master train", and all three trains were controlled based on ammonia measurements in the master train. The controller calculated DO set points for each of the three aeration zones (Pass A, Pass B, and Pass C) in each of the three trains. A parallel quad (Trains 8, 10 and 12) was instrumented identically to the controller quad and acted as a baseline for the pilot. The DO set points for the baseline trains were set by the plant operators. Figure 2 is an aerial view of the NSEC that depicts the feed-forward controller and baseline quads, and Figure 3 is a schematic of the demonstration.

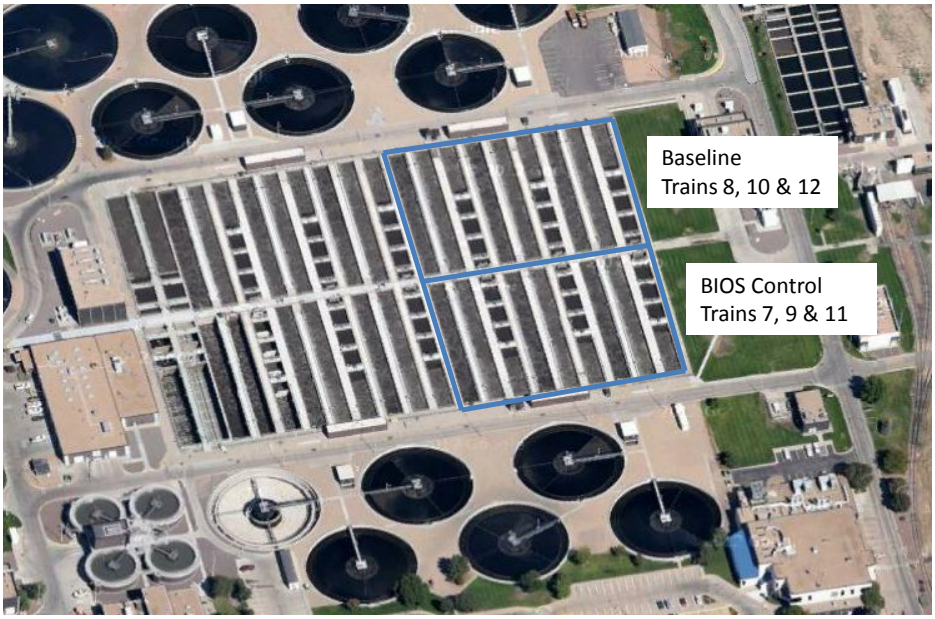

Figure 2. Aerial View of the NSEC at the RWHTF

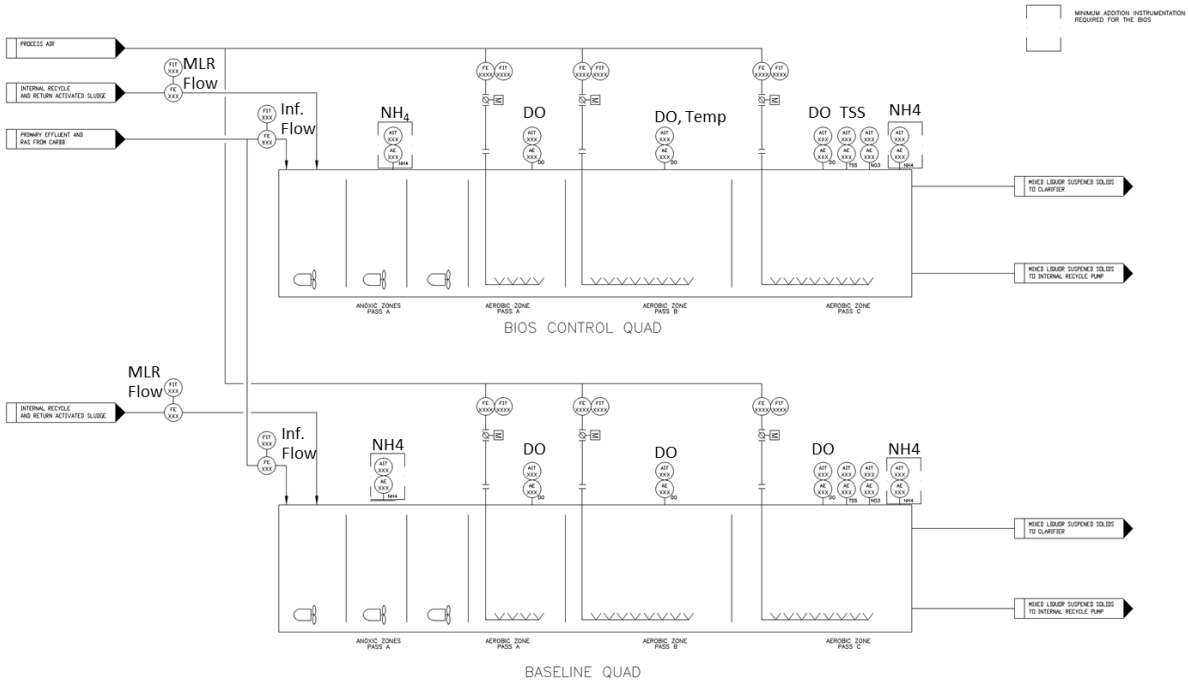

Figure 3. Process Flow Diagram of the Feed-Forward Controller Demonstration

The model only considers the aerobic section of each NSEC train. The aerobic section is modeled as eleven (11) continuously stirred tank reactors (CSTR) in series. The reactor influent (primary effluent), RAS, and mixed liquor recycle (MLR) flows are totalized and used as the flow input to the first aerobic CSTR. The output conditions of the CSTR become the input conditions of the subsequent CSTR. The model's configuration is summarized in Figure 4 below.

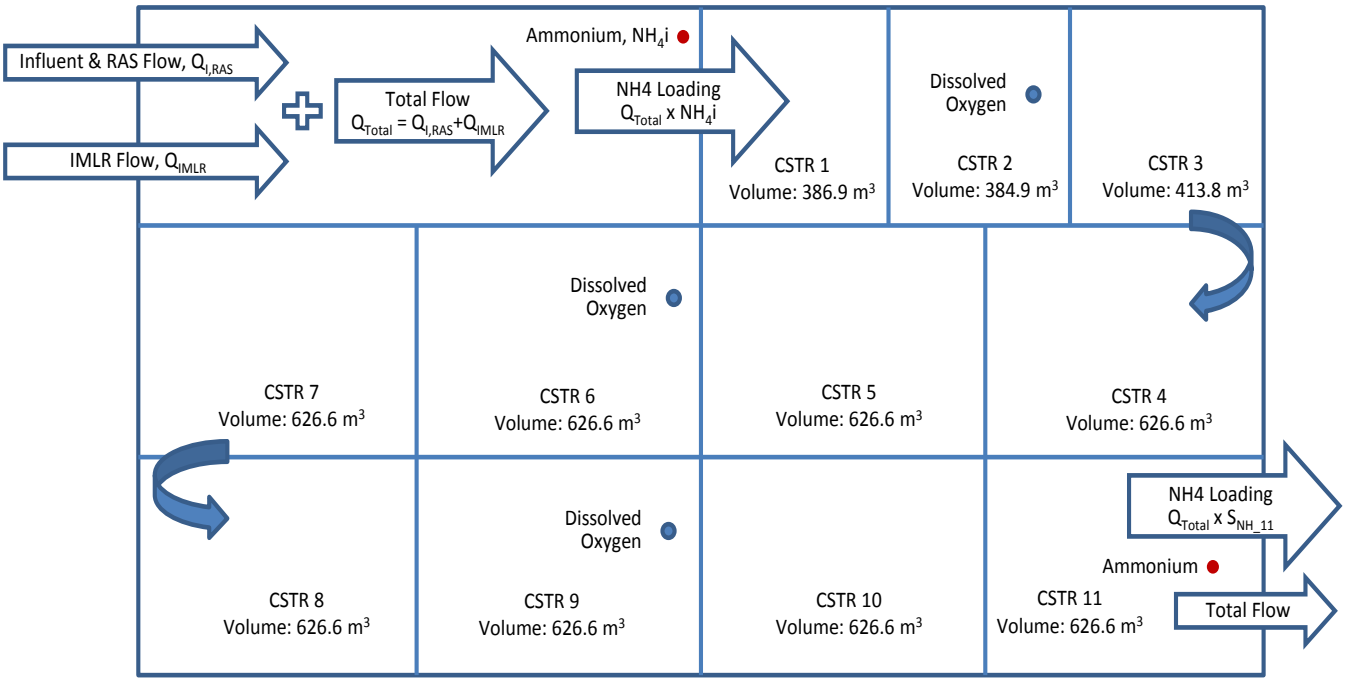

Figure 4. RWHTF NSEC Model Configuration

The controller runs on a virtual Windows 7 machine embedded in the existing Metro District server. It communicates with the plant Distributed Control System (DCS) through a Structured Query Language (SQL) Server Express database. The controller retrieves plant operating data from the database, and transmits DO set points for each aeration zone to the database for implementation by the plant DCS. For the pilot installation, the controller user interface allows operator input of set-point limits, operating data and manual overrides. The controller was installed in the second half of August, and it began operation on September 1, 2011.

NSEC Pilot Results

The controller uses a feed-forward control algorithm to provide the required DO set points to meet the variation in diurnal loadings. The aeration energy savings are generated by lowering the DO within the aeration tanks as needed by the process. As DO levels drop, the oxygen transfer efficiency increases, requiring less airflow to meet the same oxygen demand.

A typical week of DO set points for each pass of Trains 9 (Feed-forward) and 10 (baseline) is shown in Figure 5. In general the DO set points for Train 9, the feed-forward controlled train, were below the DO set points for Train 10, the baseline train. The lower DO set points resulted in decreased airflow usage, as shown on Figure 6. Despite the decreased airflow usage, nitrification performance of the two trains was similar, as shown in Figure 7. The two trains had similar influent ammonia concentrations, and similar effluent ammonia (NH_3-N) and nitrate (NO_3-N) concentrations. This indicates that the lower DO set points did not have a negative effect on nitrification performance of the secondary treatment.

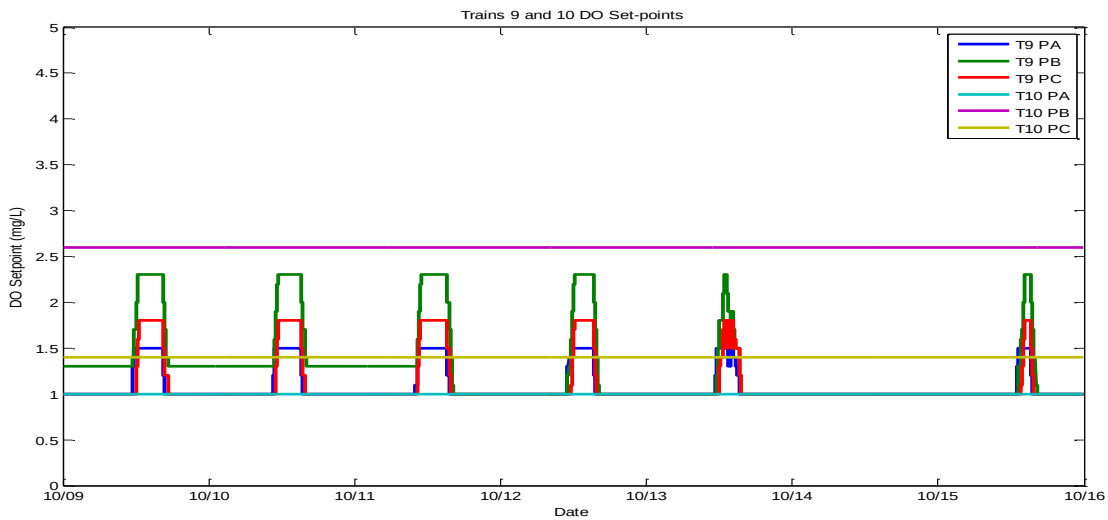

Figure 5. Train 9 and 10 DO Set points

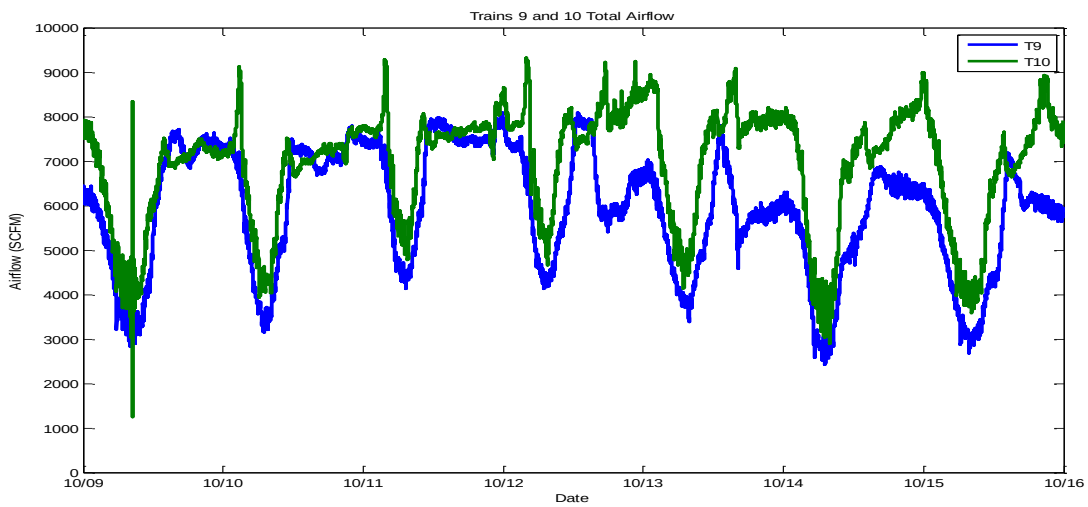

Figure 6. Train 9 and 10 Total Airflows

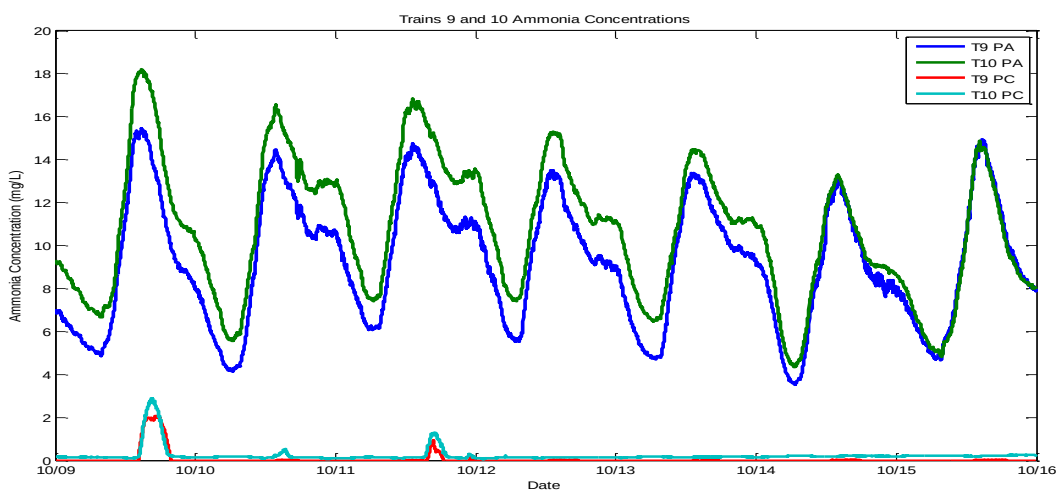

Figure 7. Train 9 and 10 Ammonia

The total airflow to each train was used to compare the feed-forward controlled quad to the baseline quad. The results of the comparisons are summarized on Figure 8. The airflow reduction between the baseline and feed-forward quads was up to 20%. During the six months of the demonstration project the BIOS controlled quad used 11% less air than the baseline quad.

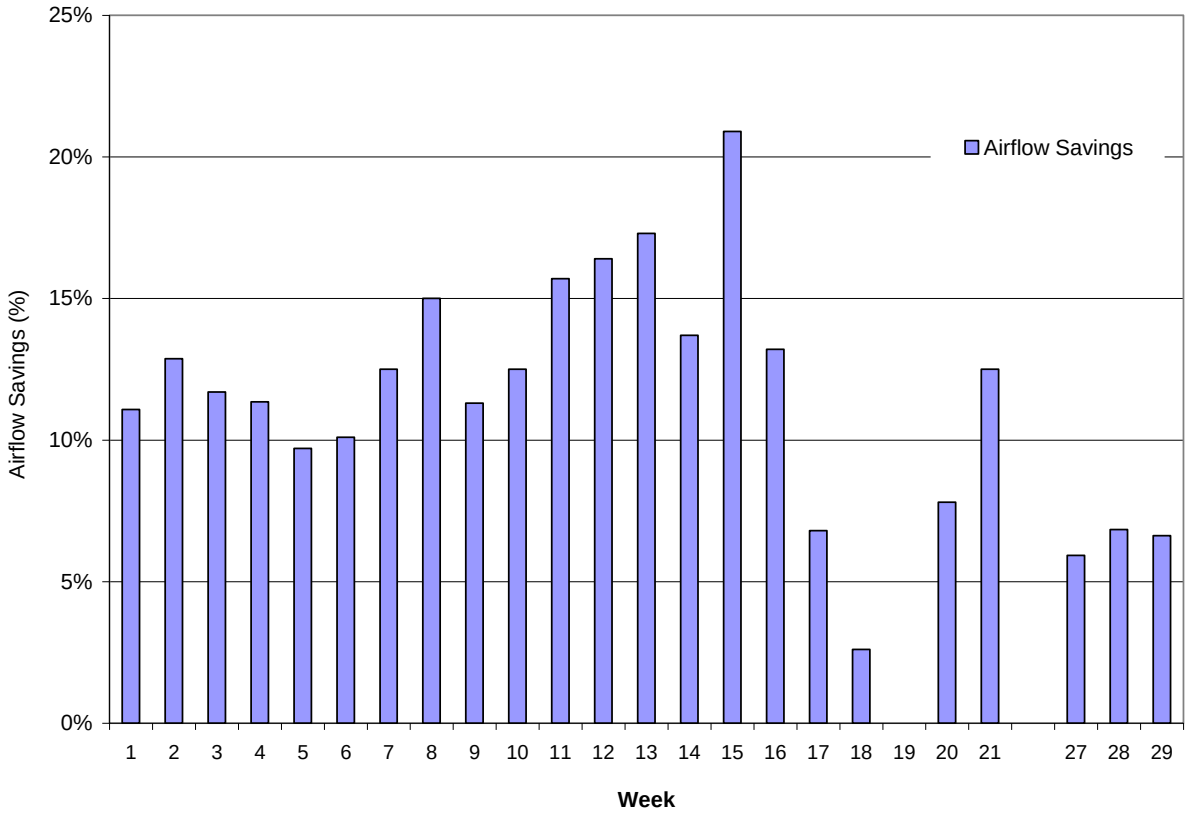

Figure 8. BIOS Aeration Savings

Prior to week 17 the average savings was 13.5%, but during week 17 the operator lowered the DO set-point for the Baseline quad and the other two quads not controlled by the controller. Because the controller DO set points were consistently lower than the baseline while meeting the effluent ammonia limit, operations felt comfortable dropping the DO set points for the rest of the plant to take advantage of the potential savings. As expected, the relative savings generated by the controller dropped, but it was still able to achieve savings of 6%. With further optimization of the selected DO range and effluent goal, the future savings can be increased. Note that the pilot was stopped during weeks 22 to 26 while the DCS system was down for maintenance and upgrades.

The feed-forward controller was part of the Metro District's NSEC Energy Optimization Initiative. Prior to the pilot installation and as part of the Initiative, the Metro District limited the number of blowers in operation to two. This limitation of blowers generated a DO sag during the peak of the day, but reduced the airflow usage by an estimated 10%. During the pilot the feed-forward controller was not affected by the limits on the number of blowers, and only required two blowers independent of the blower restriction. Comparing the DO set points calculated by

the feed-forward controller to the DO set points utilized by the plant before the NSEC Energy Optimization Initiative was started shows a theoretical savings of 22-30% using the feed-forward controller.

To place a monetary value to the observed savings, the observed airflow differentials were entered into the following equation, which includes the Metro District specific power cost in 2010 ($0.075/kWh) and conversion factor for airflow to power (0.000423 kilowatt hour per standard cubic foot [kWh/scf]):

$$\text{Daily Aeration Cost}\left(\frac{\$}{d}\right) = Airflow\left(\frac{scf}{\min}\right)\cdot 0.00423\left(\frac{kWh}{scf}\right)\cdot 1440\left(\frac{\min}{day}\right)\cdot 0.075\left(\frac{\$}{kWh}\right)$$

The estimated aeration savings during the demonstration are approximately $18,635. With a full installation of the BIOS, the estimated yearly savings would be $134,026 at the 11% observed average savings.

To fully realize the benefits of the feed-forward controller, it is essential to have an aeration system that is capable of controlling the airflow to meet the dynamic DO set points calculated by the controller. In its default setting the controller limits the DO set-point change to 0.3 mg/l per calculation cycle of 15 minutes. The NSEC aeration control system shows an acceptable level of performance for control. Aeration control over a five day period in Train 9 Pass B is shown in Figure 9.

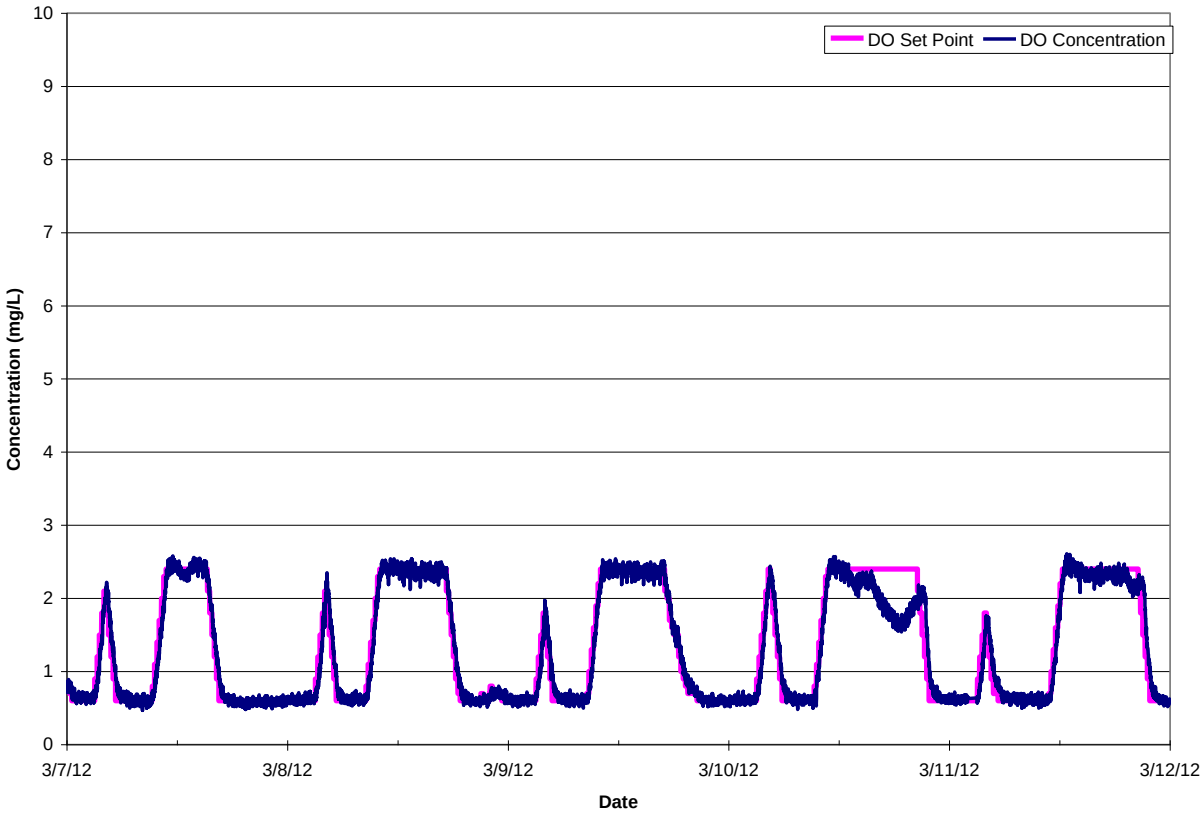

Figure 9. DO Set-Point Tracking for Train 9 Pass B

With any control system the control is only as good as the instrumentation that is used. For the feed-forward control system there are several instruments required: flow, DO, temperature, MLSS, and ammonium analyzers. Most of the instrumentation is well established within the wastewater community and requires limited maintenance. In recent years ammonium analyzers have made great strides in regards to robustness, but they still require frequent maintenance and calibration. The Metro District checked the analyzers once a week with grab samples and adjusts the analyzer calibrations curves as needed.

The Metro District provided and installed four (4) Endress & Hauser (E&H) ammonium analyzers based upon BioChem's recommendations. Overall, the ammonium analyzers worked well. Train 9 and 10 analyzer readings closely tracked each other, with only few periods of questionable readings. Figure 10 shows ammonium readings during a typical day.

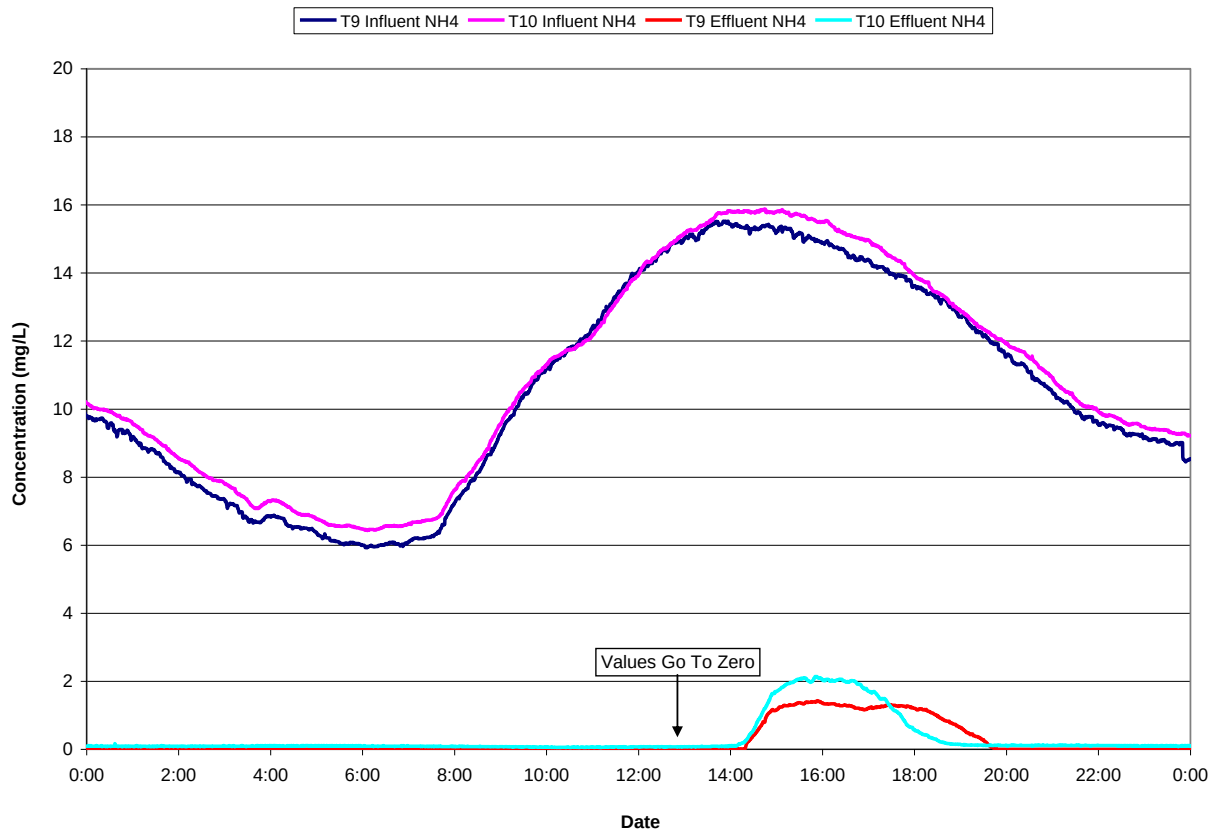

Figure 10. Train 9 and 10 Ammonium Analyzer Measurements

CASE STUDY 2

Liuwei Wastewater Treatment Plant, Yangzhou, Jiangsu Province, China

Liuwei WWTP (Liuwei) is located in Yangzhou, Jiangsu Province, China. Phase I of Liuwei has been operating since April 2005 utilizing an oxidation ditch process with a capacity of 50,000 m^3/day (13.2 MGD). By April 2007, design work for the Phase II expansion project had begun. In July 2008, the construction of the Phase II expansion began. The Phase II expansion added an A^2O process to the plant with a capacity of 100,000 m^3/day (26.4 MGD). Phase II start-up began in October 2010. In total, the Liuwei WWTP is responsible for serving an area of 19.16 square miles and a population of 390,000 persons. In order to ensure the Phase II expansion produces high quality effluent, a BIOS control system was included in the design and has been operating since February 2011.

The Phase II expansion is a completely separated process from Phase I. The liquid treatment train consists of bar screens, primary clarifiers, hydrolysis treatment, secondary treatment, secondary clarifiers, a high density sedimentation tank and microfiltration prior to final effluent discharge. The designed influent conditions and effluent permit are listed in Table 2.

Table 2: Influent Conditions and Effluent Permit for Liuwei WWTP

	COD	BOD_5	SS	TN	NH3-N	TP
Influent Conditions	450	200	230	45	35	4.0
Effluent Permit	50	10	10	15	5	0.5

The secondary treatment process consists of two trains in an anaerobic-anoxic-oxic (AAO) configuration. A diagram of the process can be found in Figure 11 showing how the flow through the major processing units.

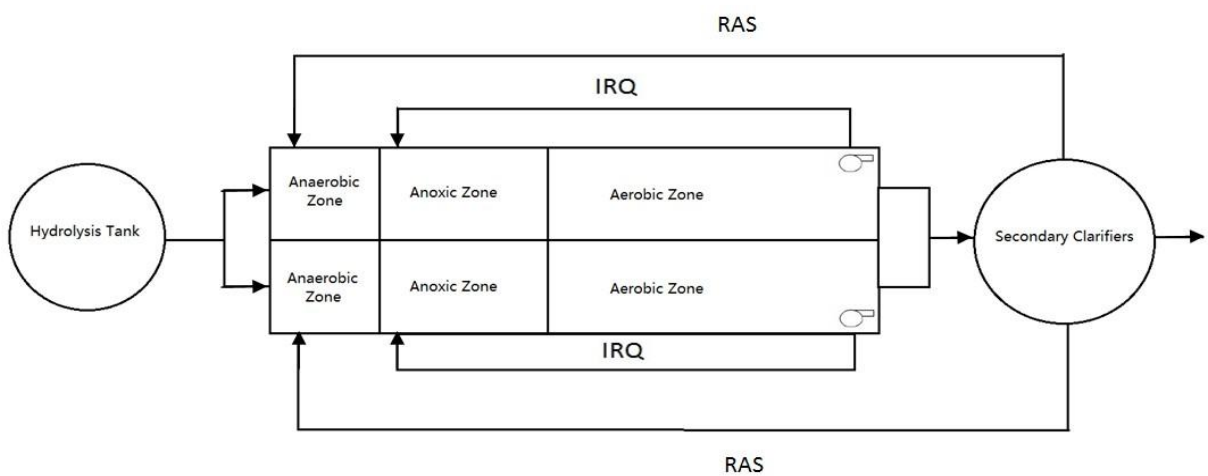

Figure 11. Major Process Units at Phase II of Liuwei WWTP

In order to implement the BIOS and maximize its benefits to the plant, installation locations of analyzers and required equipment including air valves and airflow meters were carefully selected. A diagram of one train is shown in Figure 12 indicating how these instruments were installed at Liuwei.

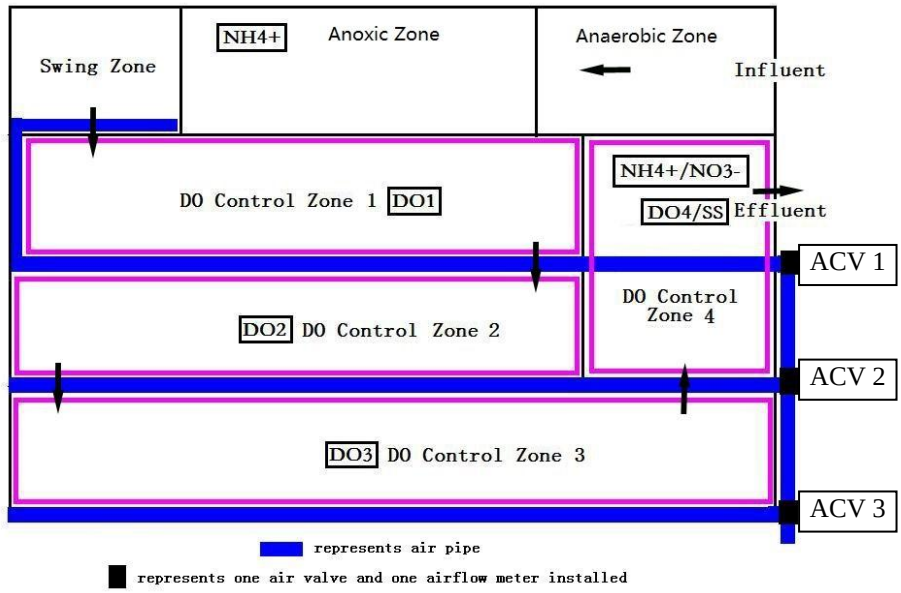

Figure 12. Layout of One Process Train and Installation Positions for Equipment

One ammonia analyzer was installed at the end of the anoxic zones and one ammonia/nitrate analyzer was installed at the end of the aerobic zones to measure the influent and effluent ammonia concentrations. For the first three DO control zones, three sets of air valves/airflow meters were installed in order to control the air entering into these zones. For DO control zone 4, manual valves were installed at each drop-leg. These manual valves are adjusted periodically to maintain a low DO in the effluent of the aeration tank in order to make sure the internal nitrate recycle doesn't return too much DO back to the anoxic zone.

Liuwei WWTP Results

The advanced control system optimizes the DO set points based on the influent flow rate and the influent ammonia concentration, as shown in Figure 13.

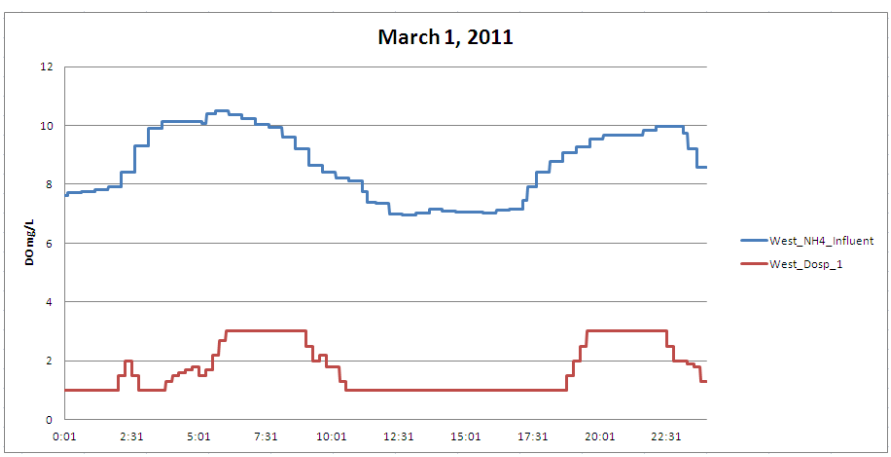

Figure 13: Calculated DO Set-Point Based On Changes in Influent Ammonia

One of the secondary objectives of implementing the advanced control system was to maintain

tight DO set-point tracking under changing influent flow rates and ammonia. Figure 14 demonstrates how the influent loading changed in the period of April 8 to 18, 2012. The incoming flow rate varied from 72,225 m^3/day (19.08 MGD) to 92,288 m^3/day (24.38 MGD) and the ammonia concentration varied from 4.7 g/m^3 to 37.2 g/m^3.

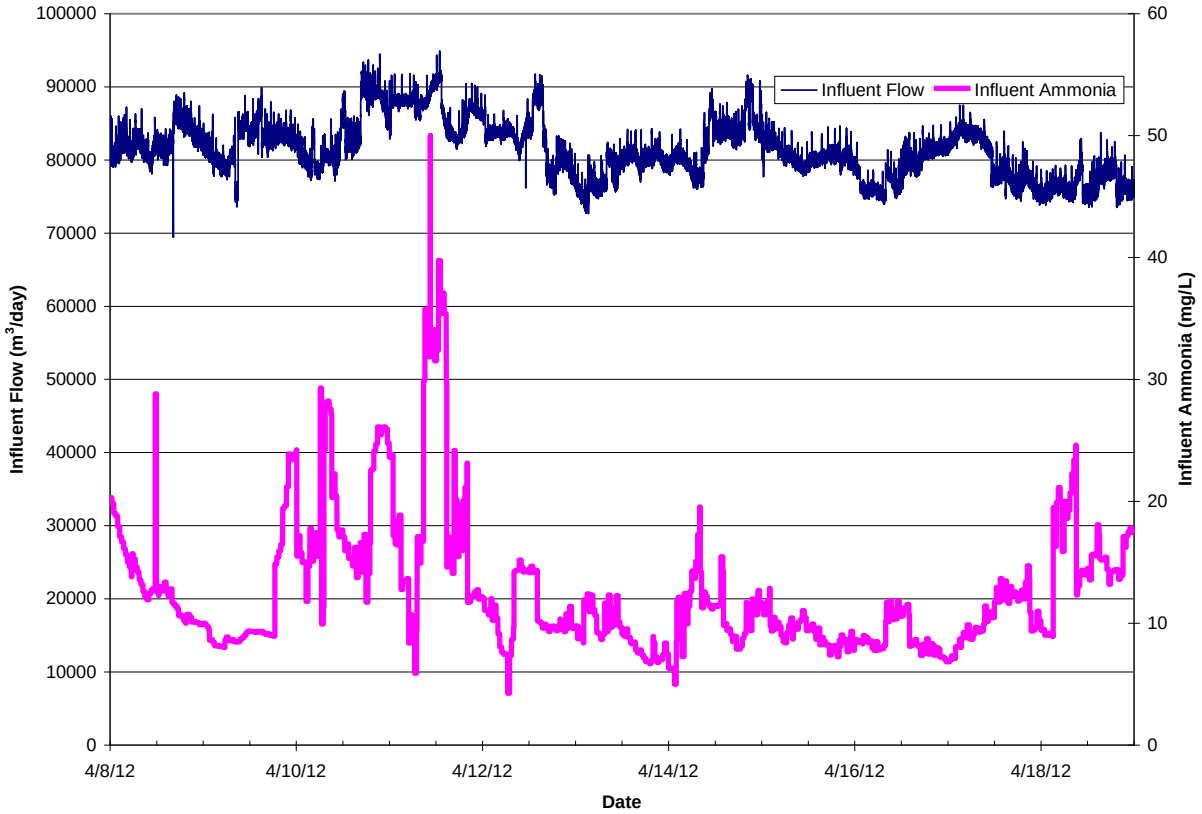

Figure 14: Changing influent flow rate and concentration over an 11-day period

DO control performance of one pass over the same period is shown in Figure 15. During this period, the DO was within ±1.0 mg/L of the set-point 99.1% of the time, and within ±0.5 mg/L of the set-point 93.9% of the time. The DO set-point was changing to match the changes in the influent loading.

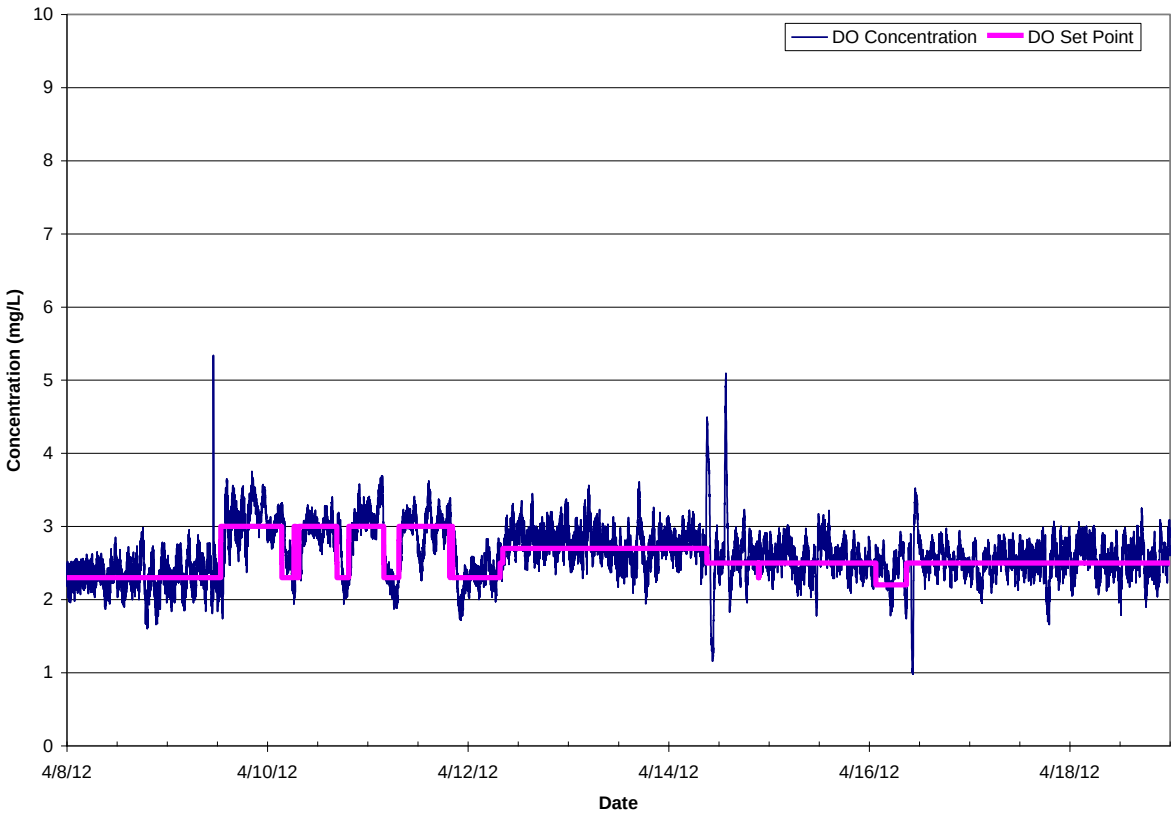

Figure 15: 11-day DO Control Curve Under Changing DO Set points

Under manual aeration control the blowers were set to supply a constant 16,000 m^3/hr of airflow which consumed an average of 195 kVA. While utilizing the advanced control system, the plant is now consuming an average of 154 kVA, an energy savings of 21.0%.

CONCLUSIONS

Overall the installations have been a success. The objective of the Metro Distric pilot project was to verify that dynamic dissolved oxygen control of the aeration system based on feed-forward control could reduce the aeration demand while meeting treatment requirements. Over the 6 months of the pilot, the feed-forward controlled quad used 12% less air than the baseline quad. With further optimization of the selected DO range and effluent goal, the savings can be increased. The estimated aeration savings during the demonstration, as a direct result of reduced airflow requirements, is approximately $18,635. With a full installation of the control system, the estimated yearly savings is $134,026. Based on these results, the Metro District proceeded with a full installation of the control system.

Likewise, the process control system at Liuwei WWTP has been a success. By implementing this control system, the plant has reduced its energy requirement by 21.0% while consistently meeting its effluent requirements.

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